

MADHAV

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EDUCATION

JSS Academy of Technical Education, Noida

Noida, India

B.Tech in Computer Science & Engineering (AI & ML) (CGPA: 7.06)

2023 – 2027

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, HTML, CSS

AI / ML & Vision: Machine Learning, Generative AI, LLMs, Computer Vision, YOLO, Face Recognition & Detection

Core CS & Algorithms: Data Structures & Algorithms, Object-Oriented Programming, Bit Manipulation, Search Algorithms

Tools & Frameworks: Full-Stack Development, API Integration, Git, GitHub, PyTorch, PettingZoo

EXPERIENCE

HCLTech

Noida, India

Software Engineering Intern

03/2026 – 05/2026

- Worked on the ETO enterprise project, delivering comprehensive technical documentation and detailed project reports aligned with organizational standards.
- Developed a real-time **Face Recognition System** using **YOLO** architecture in Python and OpenCV, optimizing detection accuracy and frame processing rates for live streams.
- Gained practical experience in enterprise software engineering workflows, cross-team collaboration, and software delivery pipelines.

MNNIT Allahabad (SNFCE)

Prayagraj, India

Research Intern

06/2026 – 07/2026

- Developed an energy-augmented Multi-Agent Reinforcement Learning (MARL) environment in PyTorch/PettingZoo based on MPE Simple-Spread.
- Implemented and benchmarked 4 reward-decomposition schemes (Uniform, Counterfactual Difference, Monte Carlo Shapley, Energy-Aware Shapley) trained under Deep Q-Networks (DQN).
- Evaluated policies across Bellman loss convergence, team reward optimization, and Jain's Fairness Index on residual battery states.

PROJECTS

🔍 **ScoutAI – AI Resume Screening & Ranking Platform** — Python, Gemini API, SQLite, Transformers

- Built a semantic search platform to rank candidate resumes against job descriptions using section-level chunking and all-MiniLM-L6-v2 embeddings.
- Implemented max-pooling across section similarity scores to prevent unrelated sections from diluting strong evidence matches.
- Integrated Gemini API with grounded prompt context to automatically generate JSON-structured gap analysis reports (strengths, gaps, and suggestions).
- Integrated public ATS job-board APIs (Greenhouse, Lever, Ashby, SmartRecruiters) and a SQLite cache with semantic role/location search capabilities.

🔍 **DriftWatch: Temporal Validation, PSI Slicing & SHAP Analysis** — Python, Streamlit, SHAP, Scikit-Learn

- Prepared public credit card fraud datasets (e.g., IEEE-CIS) by introducing synthetic time-splits to simulate production data drift and establish baseline performance.
- Developed automated tracking metrics (Precision/Recall, PR-AUC, calibration over time) and multi-segment slicing across regions/channels.
- Performed Population Stability Index (PSI) and Kolmogorov-Smirnov (KS) statistical tests to detect distribution shifts across feature slices.
- Applied SHAP-based feature attribution to analyze explainability variance between live and baseline model versions, presenting findings via an interactive Streamlit dashboard.

ACHIEVEMENTS

- Smart India Hackathon** – Participant / Qualified participant.
- AI & Machine Learning Workshop** – IIT Delhi (Tryst'24).